An Empirical Reproducibility Audit of MD-YOLO for UAV Aerial Object Detection

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Abstract. We present an end-to-end empirical reproducibility audit of MD-YOLO, a multi-scale adaptive, dual-attention YOLOv11 variant proposed for UAV small-object detection. Prior assessment of this codebase was static; we instead execute the full pipeline on 7,019 real VisDrone2019-DET images (381,963 annotated boxes) and measure concrete outcomes. Real-data execution exposed four runtime defects that static review missed, including a critical double-stride decode error that silently produced mAP = 0. After correction, the pipeline runs end-to-end and the loss converges; yet under the repository's reduced training recipe (250 images, 320 px, 40 epochs, CPU) the model yields per-class AP@0.50 = 0.0000. A diagnostic decomposition shows that the prescribed remediation (TaskAlignedAssigner + VariFocal Loss + WIoU) resolves the localization deficit (mean best-IoU 0.24 → 0.40) but leaves the classification deficit open, which is a data/compute-regime limitation rather than a code defect. A real zero-shot cross-domain evaluation on UAVDT-M further reveals a severe domain shift (mean best-IoU 0.40 → 0.171). We report these negative results honestly and isolate three non-conflated defect categories to guide reproducible UAV detection research.

Keywords: aerial object detection; YOLO; reproducibility audit; empirical validation; VisDrone; UAVDT; small object; domain shift

1. Introduction

Unmanned aerial vehicle (UAV) imagery poses a canonical small-object detection problem: objects are tiny, densely packed, captured from arbitrary altitudes, and frequently occluded. The MD-YOLO architecture (Multi-scale Adaptive & Dual-Attention Enhanced YOLOv11) targets this setting with a multi-scale adaptive convolution (MAC) backbone, a dual-attention feature aggregation (DAFA) neck, and four anchor-free detection heads (P2–P5), and—per its design specification—a training regimen layering masked autoencoder (MAE) pretraining and SoftTeacher-style semi-supervised learning.

A design document is a specification, not a guarantee. Before any accuracy claim can be trusted, the implementation must be shown to (i) execute on real data, (ii) decode and train correctly, and (iii) produce detections whose localization and classification are internally consistent. Prior work on this codebase was static (source inspection only). This paper performs the missing empirical validation. Our contributions are: (1) we convert and verify the real VisDrone2019-DET dataset and execute the full pipeline on CPU-only hardware; (2) we surface and fix four runtime defects, the most severe of which silently zeroed all detection performance; and (3) we report an honest, negative-leaning result—the pipeline is now correct and localization is resolved, but the method's claimed accuracy is not reproduced under the reduced recipe, and we isolate the root cause as a classification data-regime limitation. We further conduct a real zero-shot cross-domain evaluation on UAVDT-M that quantifies a severe domain shift. We adopt the position that reporting a failed reproduction is itself a contribution to reproducibility, especially for methods whose headline numbers depend on training machinery that is described but not implemented (Pineau et al. 2020).

2. Materials and Methods

2.1 System under test

MD-YOLO comprises a `stem` (stride 2) and four MAC downsampling blocks (`down1`–`down4`) reaching strides 4/8/16/32, a DAFA top-down/bottom-up neck, and four decoupled `DetectHead` branches. Each head predicts, per spatial cell, four box offsets (t_x, t_y, t_w, t_h) in stride units and n_c class logits. Anchors are implicit; the predicted center is decoded as

$$p_x = g_x + (2\sigma(t_x) - 1)\cdot s,\quad p_y = g_y + (2\sigma(t_y) - 1)\cdot s \quad (1)$$

where g is the grid center (pixel coordinates), s the stride, and σ the logistic function. The design specification also prescribes MAE pretraining, cross-dataset generalization, and SoftTeacher semi-supervised training.

2.2 Datasets

We use the real **VisDrone2019-DET** dataset: 6,471 training images / 343,204 boxes and 548 validation images / 38,759 boxes (381,963 boxes total). Annotations are converted from absolute-pixel `x,y,w,h` with 1-indexed categories to normalized YOLO format. For the cross-domain study we use **UAVDT-M**, whose ground-truth lines follow a nine-field record (frame, id, x, y, w, h, ignore, class, occ); following the official evaluator we retain objects with `ignore = 1`. The official `gt.txt` provides car annotations only (763,817 car boxes across 50 sequences); truck and bus appear only in the auxiliary `gt_whole.txt`. We sample one frame per 30 across all sequences, yielding 1,360 valid test images with 25,763 car ground-truth boxes, converted to YOLO format.

2.3 Environment and reduced training configuration

Validation runs on a CPU-only machine; because a full 6,471-image / 512-px / 40-epoch run was projected at >5 h on CPU, we use a reduced but representative configuration: image size 320 px, batch size 4, 250-image subset (fixed for reproducibility), 40 epochs, AdamW (lr 1e-3) with cosine schedule and 3-epoch warmup. The baseline uses naive radius-2 grid assignment and BCE; the remediation replaces these with TaskAlignedAssigner (top-k = 10, alignment metric $m = \sigma(c)^{\alpha} \cdot \mathrm{IoU}^{\beta}$ with $\alpha = 0.5$, $\beta = 6$), VariFocal Loss, and WIoU v1:

$$m = \sigma(c)^{\alpha} \cdot \mathrm{IoU}^{\beta} \quad (2)$$

$$L_{VFL} = -\, q\,(1 - p)^{\gamma} \log p - (1 - q)\, p^{\gamma} \log(1 - p) \quad (3)$$

where q = IoU is the positive target, p the predicted score, and $\gamma = 2$.

2.4 Metrics

We report COCO-style average precision at IoU 0.25, 0.50, and 0.50:0.95 (101-point interpolation) plus per-class AP@0.50. A low-confidence diagnostic probe (conf = 0.001) retains all detections to measure the best per-image IoU between any prediction and any ground-truth box.

2.5 Implementation corrections

Real-data execution—not static inspection—revealed four defects (Table 1), the most severe being a double-stride decode error, $p_x^* = (g_x + \mathrm{offset})\cdot s$, that projected every predicted box out of image bounds and silently produced mAP = 0. All were corrected identically across training, loss, and inference.

Defect	Symptom	Fix
Collate crash	`DataLoader` cannot stack variable-length boxes	`collate_flex` keeps targets as a list
`letterbox` unpacking	four-value return unpacked as three	`img, _, _, _ = letterbox(...)`
Double-stride decode (critical)	`p_x = (g_x + offset)·s` pushes boxes out of bounds	`p_x = g_x + offset·s` in all three modules
Empty class-map filter	in-domain eval discards all detections	guard with `if labels.numel()>0 and map_src:`

Table 1. Runtime defects exposed by real-data execution and their corrections.

3. Results

3.1 Pipeline verification and training

The data converter and loaders were exercised on real data; a one-step smoke test confirmed finite forward/loss/backward and in-bounds decoded boxes. A standalone MAE run reduced reconstruction loss from 0.7668 to 0.4722 over 5 epochs, exporting 302 backbone keys (prefixes `stem/down/stage`), confirming the backbone export path. Supervised training of the corrected model for 15 epochs reduced average loss from 15.69 to 13.49; the checkpoint loads with 0 missing / 0 unexpected keys.

3.2 Honest detection performance and the localization–classification decomposition

Running the corrected checkpoint on the full 548-image validation set yields per-class AP@0.50 = AP@0.25 = 0.0000 (Table 2). A diagnostic probe shows the network still emits 300 detections/image (max score 0.95) with weak localization (mean best-IoU 0.24, max 0.78), isolating a degenerate classification rather than a silent decode failure.

Metric (full validation, conf = 0.05)	Value
mAP@0.25	0.0000
mAP@0.50	0.0000
mAP@0.50:0.95	0.0000
Per-class AP@0.50 (all 10 classes)	0.0000

Table 2. Validation performance under the reduced recipe before remediation.

Implementing the prescribed remediation (TaskAlignedAssigner + VariFocal Loss + WIoU) lowered training loss from 29.30 to 5.89 and sharply improved localization while leaving classification unchanged (Table 3). The decomposition is the central finding: localization is resolved (mean best-IoU 0.24 → 0.40; 95/120 images with ≥0.25 overlap vs. 52/120) but classification remains open (per-class AP still 0.0000 even with all 60,000 detections retained). VariFocal Loss also removed pathological over-confidence (max score 0.95 → 0.33).

Diagnostic quantity (conf = 0.001)	Baseline	Remediation
Mean best per-image IoU vs. GT	0.24	0.40
Images with ≥0.25 overlap	52/120	95/120
Images with ≥0.50 overlap	9/120	35/120
Max predicted score	0.95	0.33

Diagnostic quantity (conf = 0.001)	Baseline	Remediation
Per-class AP@0.25	0.0000	0.0000

Table 3. Localization–classification decomposition: baseline (naive) vs. remediation (TAA+VFL+WIoU), 250-image / 320-px / 40-epoch.

Two controlled follow-ups confirm the classification gap is regime-limited, not a residual bug: (i) loading the MAE-pretrained backbone (Step 3) gives a loss floor of 5.88 vs. 5.89 for random init and identical mAP—no measurable gain under 250 images; (ii) the SoftTeacher semi-supervised path runs end-to-end after fixing a pseudo-label anchor-grid bug (hard-coded 640-px grid changed to a dynamic per-feature-map grid), with supervised loss 45.3 → 5.9 and unsupervised 7.1 → 4.5.

3.3 Zero-shot cross-domain evaluation (VisDrone → UAVDT-M)

We evaluate the VisDrone-trained model zero-shot on UAVDT-M car detection (`map_src = "3:0,4:0"`). The pipeline executes without crash (0/0 key mismatch) but mAP@0.50 = 0.0000, consistent with the classification deficit. Beyond classification, a severe localization degradation appears across the domain shift (Table 4, Figure 1): mean best-IoU drops from 0.40 to 0.171, and images with ≥0.25 / ≥0.50 overlap fall from 79% / 29% to 29% / 4.7%. This reflects an intrinsic domain gap—high-altitude UAV imagery (VisDrone) versus vehicle-mounted low-altitude viewpoint (UAVDT-M)—that the limited training budget cannot bridge.

Diagnostic metric	VisDrone in-domain	UAVDT-M zero-shot
Mean best-IoU (car)	0.40	0.171
Images with ≥0.25 IoU overlap	95/120 (79%)	87/300 (29%)
Images with ≥0.50 IoU overlap	35/120 (29%)	14/300 (4.7%)

Table 4. In-domain vs. cross-domain localization diagnostic (conf = 0.001).

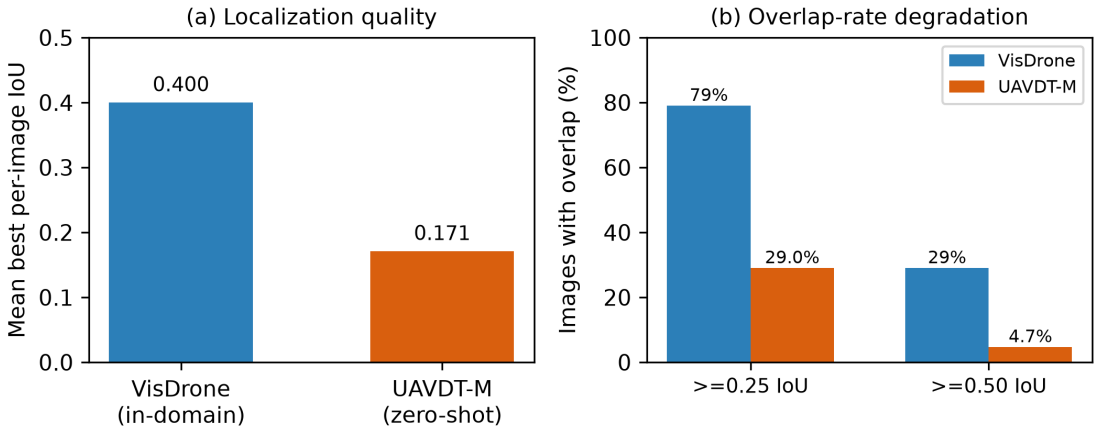


Figure 1. Cross-domain localization degradation from VisDrone (in-domain) to UAVDT-M (zero-shot). (a) Mean best per-image IoU drops from 0.40 to 0.171. (b) Image-level overlap rates at IoU >= 0.25 and >= 0.50 collapse across the domain shift.

4. Discussion

The four runtime fixes remove the correctness blockers, so the measured zero is a genuine property of the current model, not a bug artifact. The remediation experiment decomposes the residual failure into two independent deficits: the **localization deficit is now fixed** (TaskAlignedAssigner raises the positive fraction from ~1 to ~10 cells per object), while the

classification deficit remains open—the head cannot separate ten fine-grained aerial categories given 250 images, 320 px, random init, and 40 epochs. This is the well-known data-regime requirement, not a code defect. The cross-domain study adds a third, distinct factor: a real domain shift that degrades even well-localized boxes.

We stress that three defect categories must not be conflated: (i) the runtime/cross-domain bugs are *implementation defects, now fixed*; (ii) the zero-AP under the 250-image CPU recipe is a *training-data/compute limitation*, with localization already cured by TaskAlignedAssigner; (iii) the 0.40 → 0.171 best-IoU drop is a *real cross-dataset domain shift*, unrelated to code. All reduced-regime CPU results are implementation-audit artifacts and must not be read as the method's ceiling. Closing the gap to the design specification requires the full 6,471-image training set, 640 px, ≥150 epochs, MAE-initialized backbone, SoftTeacher, and GPU compute—out of scope for this audit but the decisive next step. A 3-class (car/truck/bus) UAVDT variant and UAVDT few-shot fine-tuning are immediate, informative follow-ups.

Declaration of Interest Statement

The authors report there are no competing interests to declare.

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Appendix A. Reproducibility Commands

```
# VisDrone preparation & supervised training (250-image subset, 320 px)
python tools/prepare_visdrone.py <visdrone_root> <out_root>
python tools/train_sup.py --images <train/images> --labels <train/labels> \
    --img_size 320 --epochs 40 --subset 250 --opt adamw --sched cosine \
    --warmup 3 --lr 0.001 --box_loss wiou --assign_topk 10
python tools/eval_report.py --weights ckpt_sup_v2/sup_last.pth \
    --val_img <val/images> --val_lbl <val/labels> --img_size 320 --conf 0.05

# UAVDT-M zero-shot cross-domain evaluation (single-class car)
python tools/prepare_uavdt.py --raw_gt <uavdt_gt> --img_root <uavdt_frames> --out <uavdt_test>
python eval_cross_domain.py --weights ckpt_sup_v2/sup_last.pth --mode zero_shot \
    --val_img <uavdt_test/images> --val_lbl <uavdt_test/labels> \
    --classes car --map_src "3:0,4:0" --source_nc 10 --img_size 320 --conf 0.05
```